

ML, DL, NLP Bootcamp
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Sec-2)

Introduction to
Probability

Lec-1 Probability
Addition Rule

→

Lec-2 Probability
Multiplication
Rule

• Addition Rule

- For mutually exclusive events
- For non mutually exclusive events

Multiplication Rule

- Independent & Dependent Events

Probability :-

- It is used for determining the likelihood of an event.

Ex: - Tossing of a coin

↪ It has two
Heads (H) & Tail (T)
{ 1, 2 }

$$P(H) = \frac{1}{2} = 50\%$$

$$P(T) = \frac{1}{2} = 50\%$$

Rolling a Dice
= { 1, 2, 3, 4, 5, 6 }

$$Probability (x=1) = \frac{1}{6}$$

(3)

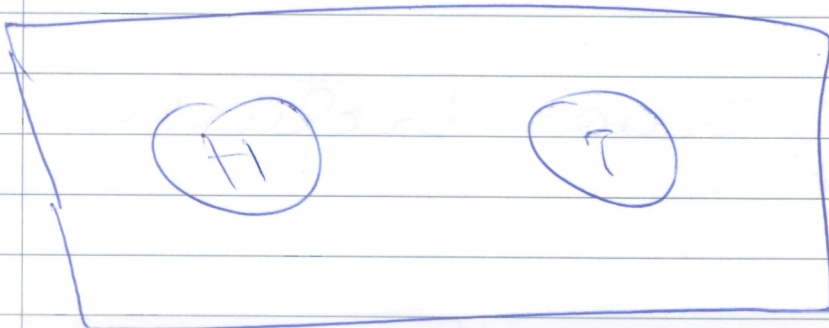
⇒ Mutual Exclusive Event

Two events are mutually exclusive if they cannot occur at the same time

Ex: - Tossing a coin

Probability

$(H \text{ or } T) =$



Probability (Head or tail)

$$= P(H) + P(T)$$

{ Adding
Rule for
mutual

Exclusive
events

$$= \frac{1}{2} + \frac{1}{2} = 1 \text{ A.}$$

(4)

Now let's take an example of Rolling a dice

Eg:- Rolling a dice
 $\{1, 2, 3, 4, 5, 6\}$

Probability (1 or 5)

$$= P(1) + P(5)$$

$$\Rightarrow \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

(2)

Non Mutual Exclusive Event:

For example if we want the probability of King or Heart King

↓
Here King is present in both the events

Hence to solve this we carry forward.

(5)

Now, we perform by
Non Mutual Exclusive
Event method i.e.

$$P(F \text{ or } C) = P(F) + P(C) - P(F \text{ and } C)$$

$\downarrow$
 Host

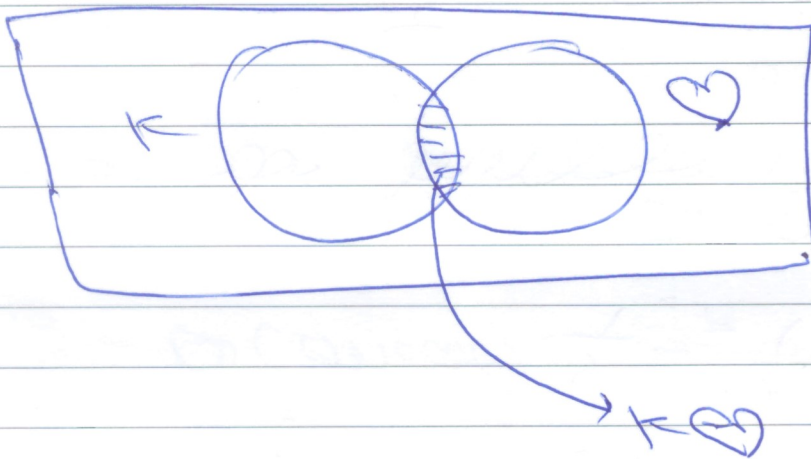
$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$P(F \text{ or } C)$$

$$= \frac{16}{52}$$

$$= \frac{1}{52} - \frac{1}{52}$$

$$= \frac{16}{52}$$



⇒ Multiplication rule
for Independent & Dependent Events

2 events are independent if they

Eg:- Tossing a coin
H & Tails

$$P(H) = \frac{1}{2}$$

$$P(T) = \frac{1}{2}$$

Eg:- Rolling of a die

$$P(1) = \frac{1}{6}$$

$$P(2) = \frac{1}{6}$$

(2)

Independent events

- 2 events are independent when the effect of one does not affect the other.

Ex: - There is a bag with 10 balls, 5 red and 5 blue. Two balls are drawn without replacement.

$$P(E) = \frac{4}{92} \rightarrow \text{probability of getting 4}$$

→ If one ball is taken out, the probability of getting the other ball is 1/91.

$$P(\text{success}) = \frac{4}{91} \rightarrow \text{4 balls left}$$

(1)

Independent Event
of Tossing a coin

$$P_2(\text{Head and Tail}) = P_2(H) * P_2(T) \\ = \frac{1}{2} * \frac{1}{2} \\ = \frac{1}{4}$$

(2)

Four Dependent Events, and
take forward as:

$$P_2(K \text{ and } Q) = P(K) * P(Q|K) \\ = \frac{4}{52} * \frac{4}{51}$$

Probability of Q
given K that is dependent
conditional
Probability